Simulation data-generating models

Let $X = (X_1, \ldots, X_p)$, where the components are independent Uniform$(-1, 1)$, let $U \sim$ Uniform$(-1, 1)$, and let $Z^* \in \{-1, 1\}$ be randomised with probability one half. The observed treatment and outcome are binary. The scripts recode binary variables to $\{-1, 1\}$ before fitting.

Main simulation

For the main setting, $p = 5$, $\lambda = \xi = \delta = \eta$, and

$$A \mid X, U, Z^* \sim \text{Bernoulli}\{\text{expit}(8Z^* + X_1 - 7X_2 + \lambda(1 + X_1)U)\}.$$

Conditional on (X, U, A),

$$Y \sim \text{Bernoulli}\{\text{expit}(g_1(X, U) + g_2(X, U)A)\},$$

where

$$g_1(X, U) = \begin{cases} 1 - X_1 - X_2 + \xi U, & \text{setting } 1 = 1, \\ 1 - X_1^2 + X_2^2 + \xi X_1 X_2 U, & \text{setting } 1 = 2, \end{cases}$$

and

$$g_2(X, U) = \begin{cases} 0.442(1 - X_1 - X_2 + \delta U), & \text{setting } 2 = 1, \\ X_2 - 0.25\delta X_1^2 + 4I(X_2 > 0) - 6I(X_2 \leq 0), & \text{setting } 2 = 2. \end{cases}$$

The main grid uses $n \in \{300, 500\}$, $\eta \in \{0.5, 1.5\}$, both setting indicators in $\{1, 2\}$, and $d \in \{0.5, 0.4, 0.3, 0.2, 0.1\}$.

Additional simulations under varying design features

These additional simulations retain the same outcome model and vary one first-stage feature at a time: IV strength, covariate dimension, or the strength of unmeasured confounding in treatment assignment. They otherwise use setting $(2, 2)$ with $\eta = 1.5$. The mild setting removes the U-to-A term and uses $\text{logit}\, P(A = 1 \mid Z^*, X) = 2Z^* + X_1 - X_2$. See `test_variant.R` and `test_mild.R` for the exact coefficients and grids.

Identifiable setting

The identifiable setting makes the minimal first-stage change of removing the unmeasured confounder from treatment assignment while retaining the main manuscript first-stage coefficients:

$$A \mid X, Z^* \sim \text{Bernoulli}\{\text{expit}(8Z^* + X_1 - 7X_2)\}.$$

The outcome model and the four combinations of settings 1 and 2 are exactly as above, with $p = 5$ and $\xi = \delta = \eta$. Consequently the conditional treatment effect is identified by the Wald-type comparison without deferral. The proposed method is still evaluated over the same five values of d.

For this setting, the scripts report both the worst-case loss computed from the population partial-identification bounds and the loss using the true CATE. The code checks at every replication that the former is no smaller than the latter under the implemented reparameterisation of d.